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REPORT

A STAKEHOLDER-CENTERED MINI REVIEW OF
FAIRNESS IN MUSIC RECOMMENDER SYSTEMS

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**REPORT ON: A Stakeholder-Centered Mini
Review of Fairness in Music Recommender
Systems**

**BASED ON THE PAPER: “Fairness in Music
Recommender Systems: A Stakeholder-Centered
Mini Review”**

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1. Bibliographic Information

Title of the paper: “Fairness in Music Recommender Systems: A Stakeholder-Centered Mini Review”

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2. Research Objectives and Aims

The paper aims to “take stock” of the current state of research on fairness in Music Recommender Systems (MRS).

The main research question is: “What is the state of the art of **MRS** fairness research from the various stakeholder’s perspectives?”

The goal is to synthesise and structure the literature to identify key findings, trends, research gaps and to understand what is known about fairness for different affected parties and where the research people needs to focus their efforts next.

3. Methodology

Research Method: The paper used a qualitative, literature-based methodology. It is a **Mini Review**.

Data Collection: The researcher considered various attributes like **Popularity, User country, age, gender, Artist gender, genre** etc.

Specific Methodology: The author's main analytical tool was the “**stakeholder-centred framework**” which they used to categorise and analyse the literature. They grouped the findings based on which stakeholder is the focus of the fairness concern.

Data Sources: The data has been collected from various Datasets including **LFM-1b, LFM-1K, Last.fm, Spotify, LFM-360K, FM4 SoundPark** etc.

4. Theoretical Framework

The research papers's complete analysis was built on a stakeholder-centred theoretical framework. The framework states that “fairness” is not a single, rigid concept but it is defined differently by the various researchers involved in the MRS ecosystem.

The Stakeholders are defined as:

- **Users:** The listeners. For them, fairness is to receive a comparable quality of service and recommendation experience regardless of their personal tastes in music.
- **Artists:** The creators. To them, fairness is to get equal exposure. The algorithm should not hide them based on their popularity, gender or location.
- **The Platform:** The intermediary. This stakeholder is responsible for defining and implementing fairness constraints.

5. Key Findings

- **User's Perspective:**
 - **Popularity Bias:** Users with “mainstream” tastes gets better recommendations compared to others. Users with “niche” tastes gets poor recommendations.
 - **Other Biases:** The research showed that users from minority gender or location receives worse recommendations.
- **Artist's Perspective:**
 - **Bias Amplification:** MRS algorithms shows social biases and amplify them. Means popular artists gets more exposure than rising artists.
 - **Locality and Popularity Bias:** Artist outside the big market like US/UK and rising artists gets overshadowed by the algorithm that focuses on popular content and higher clicks.
- **The Main Finding:** The majority of the researches only focused on identifying and analysing the problem but very few actually provided any sort of solution or algorithm to fix it.

The findings supports the paper's subtle hypothesis that the field of MRS fairness is much focused on problem analysis and not on problem solving.

6. Contributions

The research paper's main contribution is not a new algorithm or solution but **a synthesis and critique of the research field**. It clearly points out the major gap in “analysis” and “mitigation”. It asks researchers to shift their focus to provide a solution rather than analysing it.

It provides a simple and effective stakeholder-centred framework to organise and understand the complex and biased nature of fairness in music recommendations.

It filled a gap by providing a concise and updated summary of the field making it easier to access by new researchers.

7. Limitations

- **Scope:** As a “mini-review” the paper does not provide a complete review of all possible literature. It just provides a summary.
- **Field Limitations:** There’s not quite such limitations found from the review. Rather mostly limitations are from the field the paper is reviewing.
- **Observed Limitations:** The paper follows a strong organising principle but it is seen few other stakeholders like **songwriters and record labels** are almost completely ignored.

8. Literature Review Context

This paper is a literature review, so its entire context is the main work it is reviewing. It works on finding the biases in recommendations systems in general and applies it to music.

The key studies are:

- Identify popularity bias.
- Identify gender and location biases in music datasets and algorithms.
- Interview artists about their fairness concerns.

The paper does not just present one finding at a time but organises all the findings to put a statement about the state of the field itself.

9. Discussion and Implications

- **Author's Interpretation:** The authors interpret their findings to mean that the field of MRS fairness is still in growing phase. The researchers showed that bias exist but provided no notable solution to overcome them.
- **Practical Implications:** The implications for platforms is that they can't rely on academic research for solutions only. They must invest rather in in-house R&D to address these issues.
- **Theoretical Implications:** The research paper shows that future researches should focus on multi-stakeholder perspective as one can affect other.
- **Future Research:** The paper directly tells researchers to shift their research towards designing and developing an algorithm to solve the bias problem.

10. Research Gaps

The paper is quite clear about the research gaps:

1. **The Analysis-Mitigation Gap:** The most significant gap is the “lack of approaches that improve the fairness”.
2. **Data Scarcity Gap:** This field suffers from lack of large, public, high quality datasets that contains meaningful datas.
3. **Missing Stakeholder Gap:** Various important stakeholders are overlooked like songwriters, composers, record labels etc.

11. Conclusion and Future Suggestions

The paper concludes with the fact that even if researchers agree with the existence of multiple biases the progress in this field is still slow. The lack of resources and research on algorithms stopping to progress in solving the issues of biases.

The primary future suggestion is researchers must focus on developing a solution rather than just analysing it. Also propose new metrics to evaluate that account for multi-stakeholder fairness and working to create a better representative datasets for the community.

12. Summary

Methodology	Dataset	Results Obtained	Key Findings	Drawbacks / Research Gaps
Literature review using a Stakeholder-Centered Framework to synthesise existing research.	No new data was collected.	A summary of the overall research field. The main result is that the field is focused on analysis and not on solution.	1. Biases Confirmed - Popularity, gender and locality affect both users and artists. 2. Bias Amplification - Algorithms promotes the social bias. 3. Multi-Stakeholder research is nil.	1. The Analysis-Mitigation Gap. 2. Lack of Data. 3. Missing Stakeholders.

END
